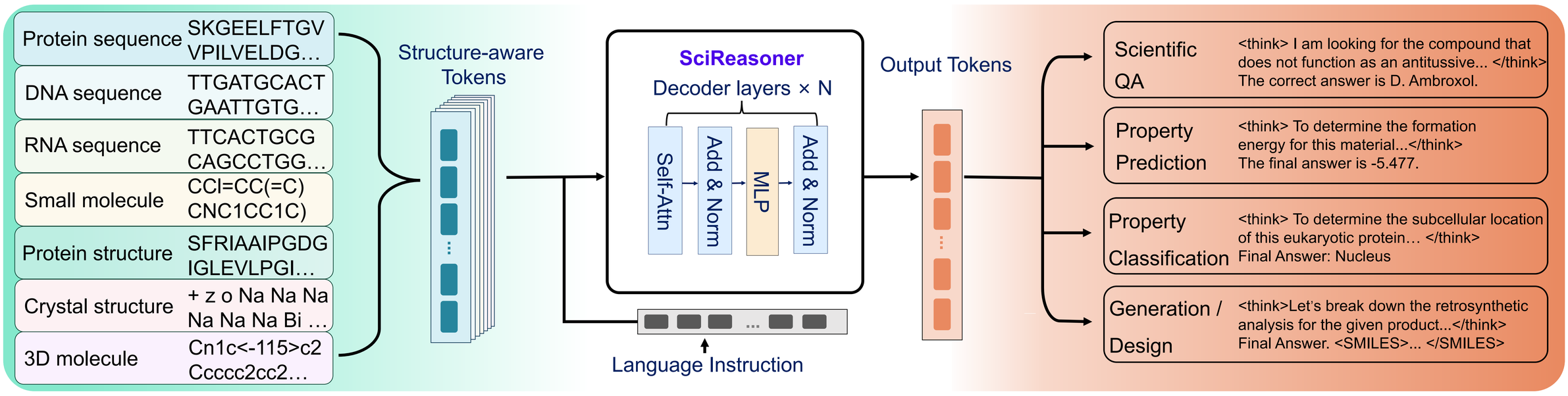
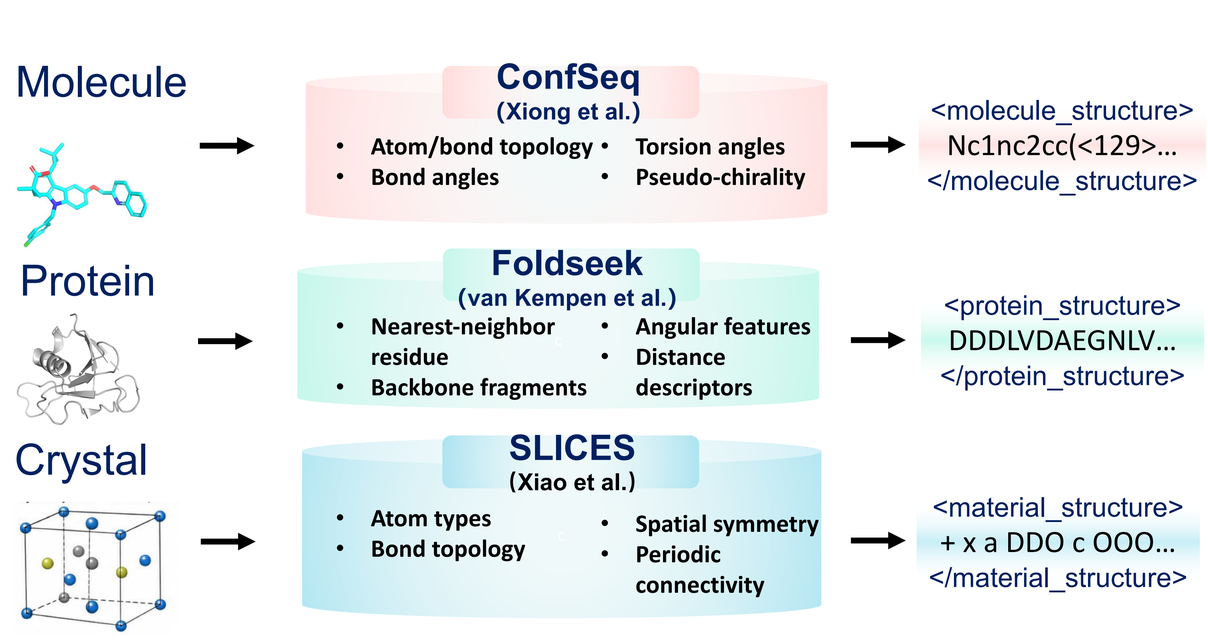
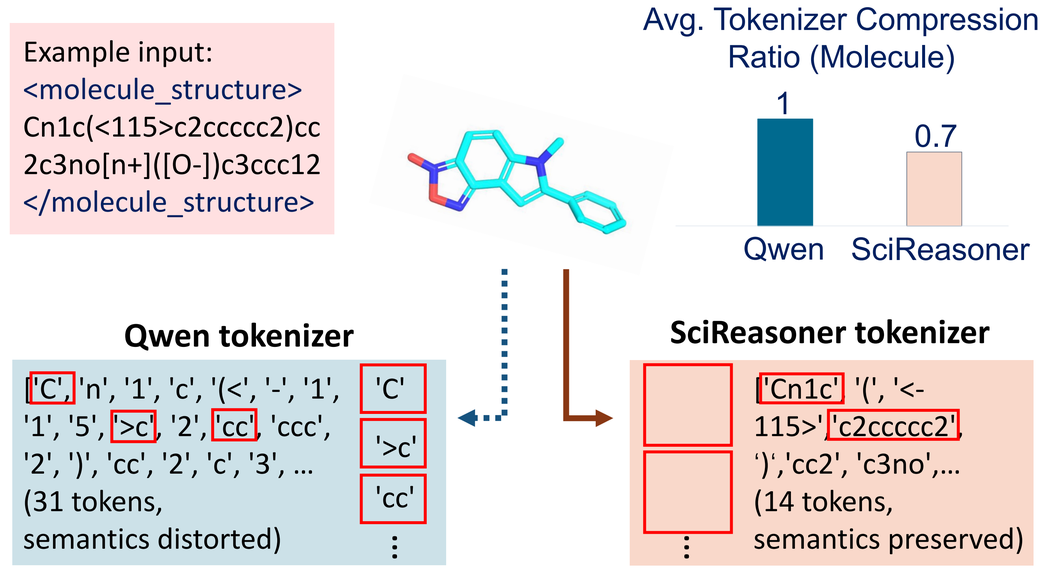
A


Protein sequence
SKGEELFTGV VPILVELDG…
DNA sequence
TTGATGCACT GAATTGTG…
RNA sequence
TTCACTGCG CAGCCTGG…
Small molecule
CCl=CC(=C) CNC1CC1C)
Protein structure
SFRIAAIPGDG IGLEVLPGI…
Crystal structure
+ z o Na Na Na Na Na Na Bi …
3D molecule
Cn1c<-115>c2 Ccccc2cc2…
Structure-aware Tokens
SciReasoner
Decoder layers × N
Self-Attn
Add & Norm
MLP
Add & Norm
Language Instruction
Output Tokens
Scientific QA
<think> I am looking for the compound that does not function as an antitussive... </think>
The correct answer is D. Ambroxol.
Property Prediction
<think> To determine the formation energy for this material...</think>
The final answer is -5.477.
Property Classification
<think> To determine the subcellular location of this eukaryotic protein… </think>
Final Answer: Nucleus
Generation / Design
<think>Let's break down the retrosynthetic analysis for the given product...</think>
Final Answer. <SMILES>... </SMILES>

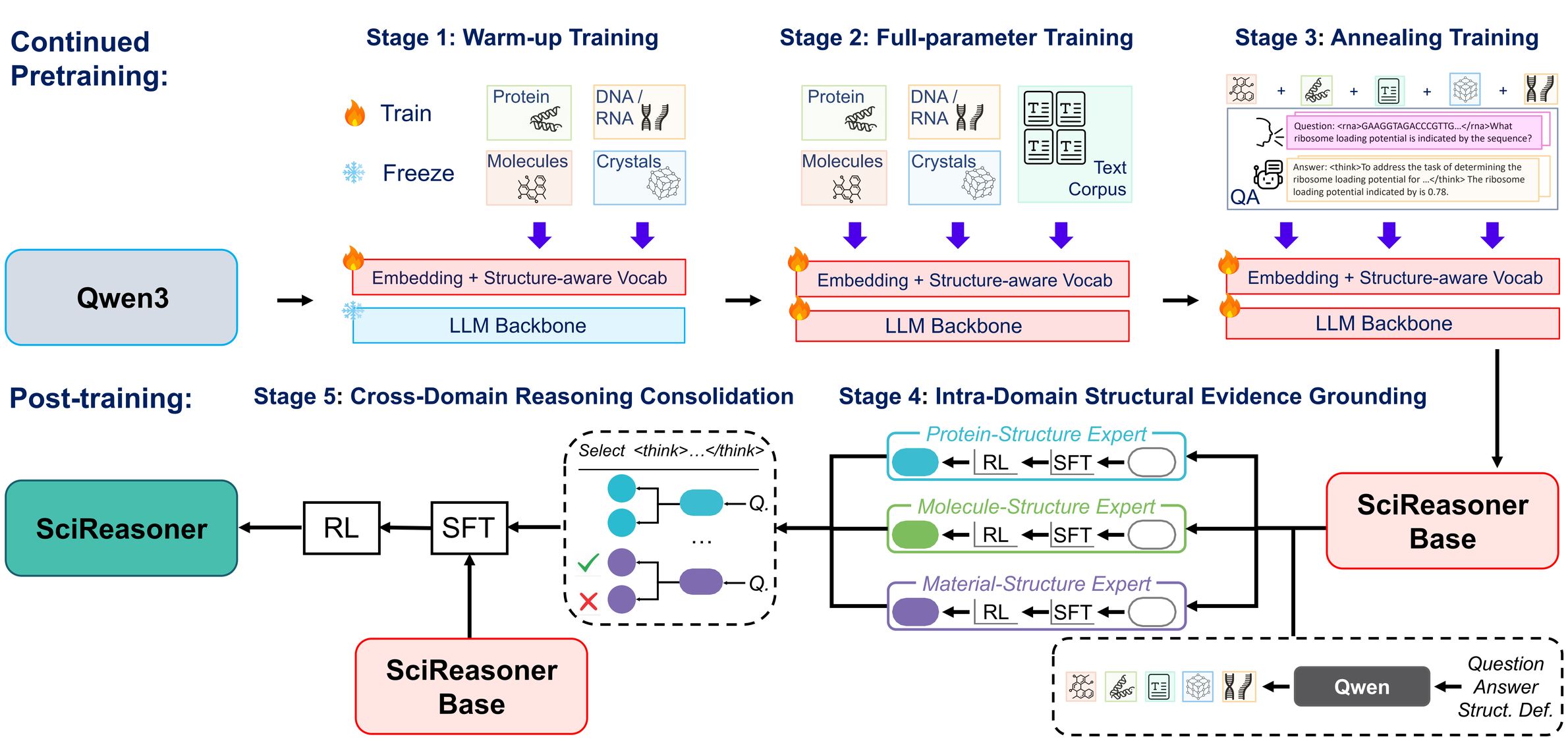
B


Molecule
Protein
Crystal
ConfSeq
(Xiong et al.)
Atom/bond topology
Bond angles
Torsion angles
Pseudo-chirality
Foldseek
(van Kempen et al.)
Nearest-neighbor residue
Backbone fragments
Angular features
Distance descriptors
SLICES
(Xiao et al.)
Atom types
Bond topology
Spatial symmetry
Periodic connectivity
<molecule_structure> Nc1nc2cc(<129>... </molecule_structure>
<protein_structure> DDDLVDAEGNLV... </protein_structure>
<material_structure> + x a DDO c OOO... </material_structure>

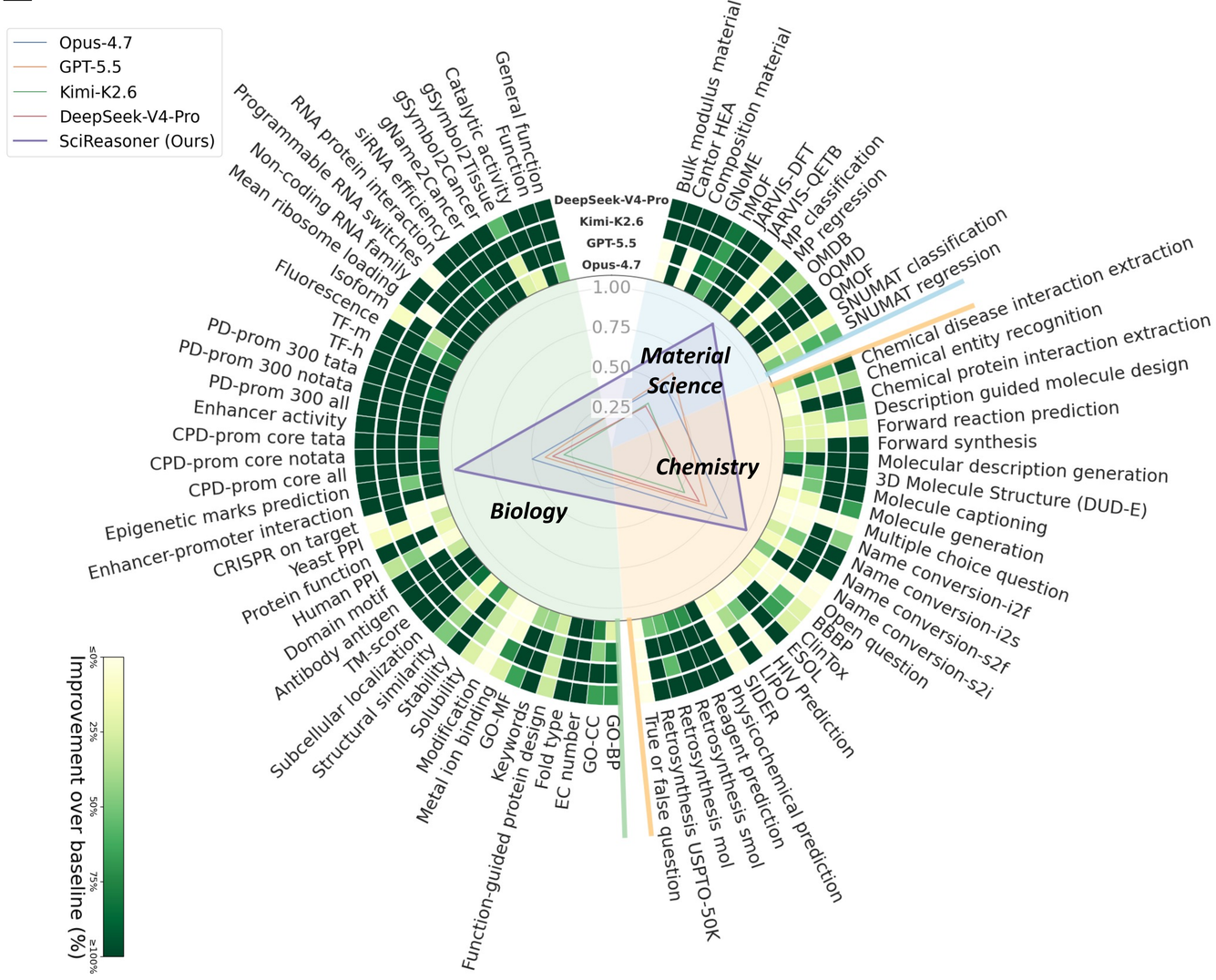
C

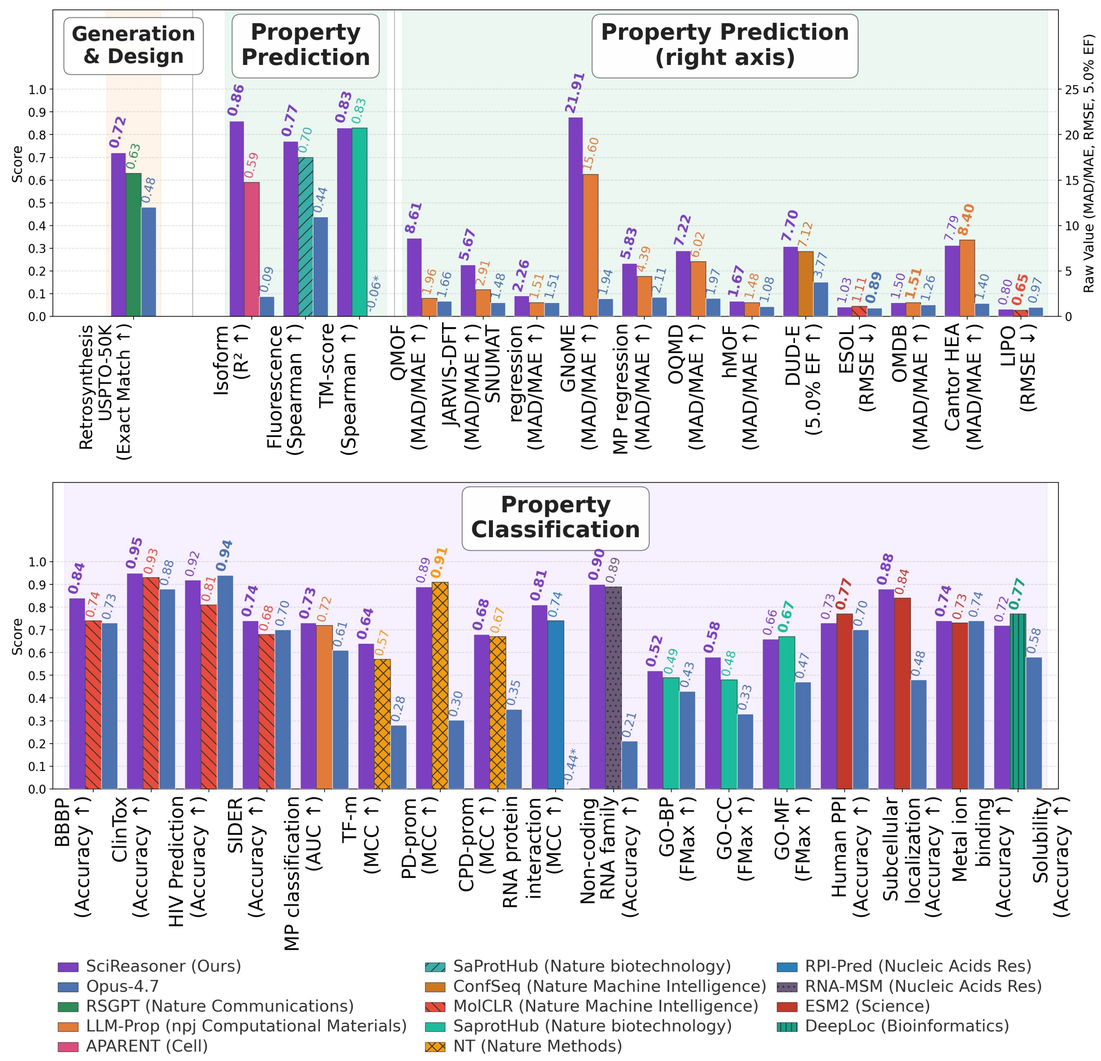

Example input:
<molecule_structure>
Cn1c(<115>c2ccccc2)cc
2c3no[n+]([O-])c3ccc12
</molecule_structure>
Avg. Tokenizer Compression Ratio (Molecule)
1
0.7
Qwen
SciReasoner
Qwen tokenizer
['C', 'n', '1', 'c', '(<', '-', '1', '1', '5', '>c', '2', 'cc', 'ccc', '2', ')', 'cc', '2', 'c', '3', …
(31 tokens, semantics distorted)
'C'
'>c'
'cc'
SciReasoner tokenizer
['Cn1c' '(', '<-115>' 'c2ccccc2', ')','cc2', 'c3no',…
(14 tokens, semantics preserved)

D


Continued Pretraining:
Stage 1: Warm-up Training
Stage 2: Full-parameter Training
Stage 3: Annealing Training
Train
Freeze
Protein
DNA / RNA
Molecules
Crystals
Text Corpus
Question: <rna>GAAGGTAGACCCGTTG…</rna>What ribosome loading potential is indicated by the sequence?
Answer: <think>To address the task of determining the ribosome loading potential for …</think> The ribosome loading potential indicated by is 0.78.
QA
Qwen3
Embedding + Structure-aware Vocab
LLM Backbone
Post-training:
Stage 5: Cross-Domain Reasoning Consolidation
Stage 4: Intra-Domain Structural Evidence Grounding
SciReasoner
RL
SFT
Select <think>…</think>
Q.
SciReasoner Base
Protein-Structure Expert
Molecule-Structure Expert
Material-Structure Expert
SciReasoner Base
Qwen
Question
Answer
Struct. Def.

E


Opus-4.7
GPT-5.5
Kimi-K2.6
DeepSeek-V4-Pro
SciReasoner (Ours)
Material Science
Chemistry
Biology
Improvement over baseline (%)


Generation & Design
Property Prediction
Property Prediction (right axis)
Property Classification
Score
Raw Value (MAD/MAE, RMSE, 5.0% EF)
SciReasoner (Ours)
Opus-4.7
RSGPT (Nature Communications)
LLM-Prop (npj Computational Materials)
APARENT (Cell)
SaProtHub (Nature biotechnology)
ConfSeq (Nature Machine Intelligence)
MolCLR (Nature Machine Intelligence)
SaprotHub (Nature biotechnology)
NT (Nature Methods)
RPI-Pred (Nucleic Acids Res)
RNA-MSM (Nucleic Acids Res)
ESM2 (Science)
DeepLoc (Bioinformatics)